3BLUE1BROWN · ESSENCE OF CALCULUS

Calculus
Complete Notebook

A complete study and revision reference for the Calculus step of the AI Research Roadmap. Every required chapter, rewritten as lessons with worked calculations.

What is inside

7 chapters	Chapters 1–7, in order
Full lessons	Plain-language explanation of every concept
Diagrams	Geometric pictures for the visual ideas
Worked examples	Step-by-step differentiation, fully written out
Tests + answers	3 questions per chapter, easy · normal · hard
Formula boxes	Key rules collected at the end of each chapter
Master reference	All formulas in one table + verify habits

Chapters 8–12 (integration, Taylor series) are skipped — not required at this stage. The course stops after the chain rule and limits.

AI Research Roadmap · Phase 1 · Stage 1 · Step 2

The essence of calculus

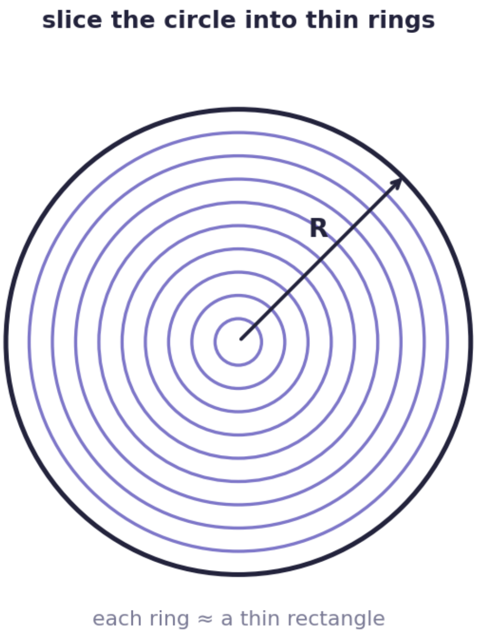
Three big ideas, discovered from one circle.

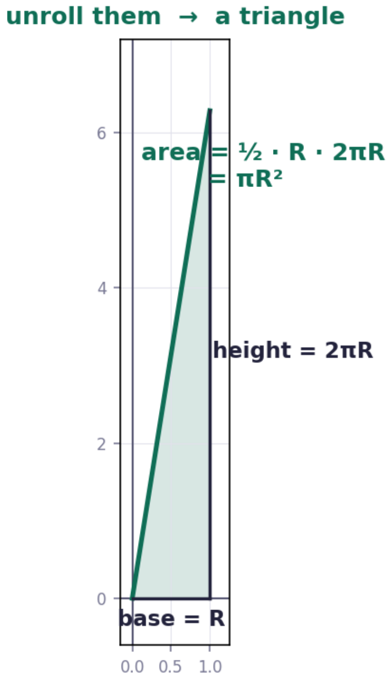
THE IDEA

Calculus is not a pile of formulas to memorise — it is three connected ideas, and you can stumble into all three by asking one simple question: **why is the area of a circle πR^2?**

THE CIRCLE ARGUMENT

Slice the circle into very thin rings. One ring at radius r has thickness dr and length $2\pi r$, so its area is roughly $\mathbf{2\pi r \cdot dr}$. Now unroll all the rings and stand them side by side: they form a triangle with base R and height $2\pi R$. Its area is $\frac{1}{2} \cdot R \cdot 2\pi R = \mathbf{\pi R^2}$.





THE THREE IDEAS HIDING IN THAT ARGUMENT

Idea	What it is
Integral	Adding up infinitely many tiny pieces to get a total
Derivative	How fast a quantity is changing at an instant
They are opposites	Integrating and differentiating undo each other — the fundamental theorem

The strategy behind all of calculus: **approximate a hard problem with many easy pieces, then make the pieces infinitely small.**

WORKED EXAMPLE

CHAPTER 1 — WORKED EXAMPLE

the three big ideas, seen in one picture

THE CIRCLE

one ring at radius r : thickness dr, length $2\pi r$

ring area $\approx 2\pi r \cdot dr$

ADD ALL THE RINGS (this is an INTEGRAL)

stack them → a triangle with base R and height $2\pi R$

area $= \tfrac{1}{2} \cdot R \cdot 2\pi R = \pi R^2$

THE THREE IDEAS

Integral	adding up infinitely many tiny pieces
Derivative	how fast a quantity changes
They are opposites	the fundamental theorem of calculus

Why it matters for ML: training a model means asking "if I nudge this weight slightly, how much does the loss change?" That is a derivative — and every weight in the network has one.

FORMULAS FROM THIS CHAPTER

Ring area	$2\pi r \cdot dr$
Circle area	πR^2
Integral	add up infinitely many tiny pieces
Derivative	rate of change at an instant

TEST — Chapter 1

1. A ring at radius $r = 3$ has thickness $dr = 0.1$. Estimate its area using $2\pi r \cdot dr$. **→ $2\pi(3)(0.1) \approx 1.88$**
2. Using the triangle picture, what is the area of a circle with $R = 4$? **→ $\tfrac{1}{2} \cdot 4 \cdot 2\pi(4) = 16\pi \approx 50.3$**
3. In one word each: what does an integral do, and what does a derivative do? **→ integral: accumulates · derivative: measures change**

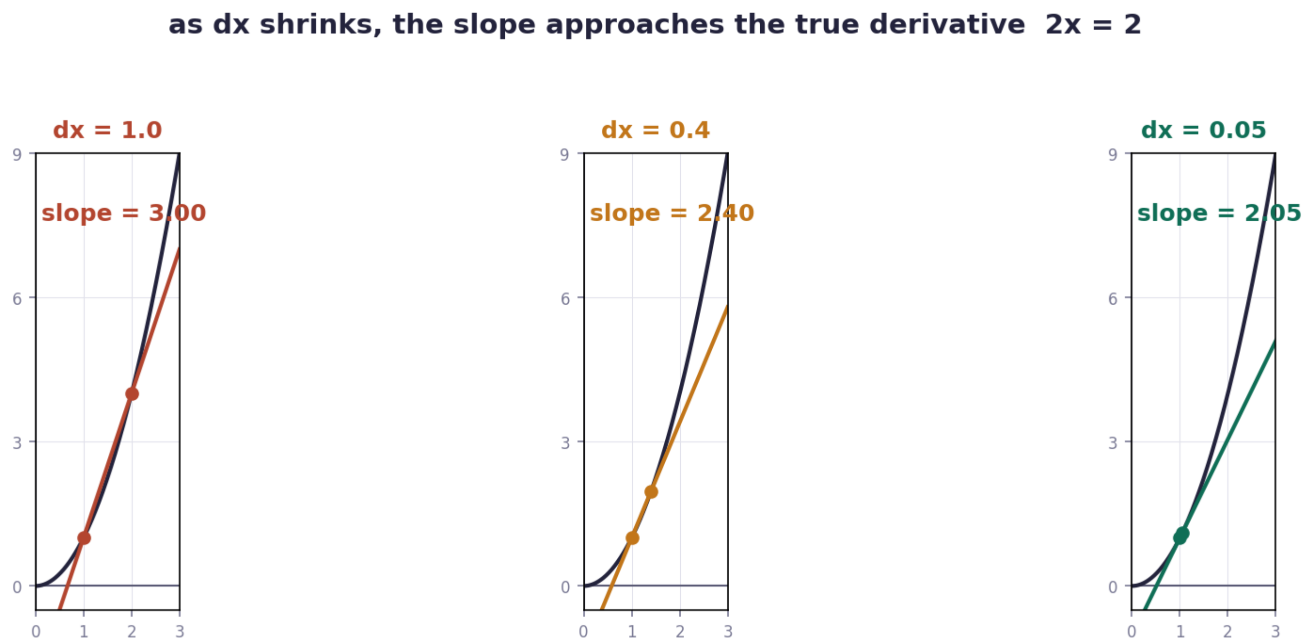
The paradox of the derivative

Rate of change at a single instant.

THE IDEA

A derivative measures how fast something changes *at one instant* — but change needs two points to measure. That is the paradox. The resolution: take two points a distance dx apart, compute the slope, then ask what that slope **approaches** as dx shrinks toward zero.

> **The definition:** $df/dx = [\ f(x + dx) - f(x)\ ] / dx$, as $dx \to 0$.



WHAT THE DERIVATIVE TELLS YOU

Sign of df/dx	Meaning
positive	The function is increasing
negative	The function is decreasing
zero	Flat — a peak, a valley, or a plateau

Geometrically it is the **slope of the tangent line** at that point.

WORKED EXAMPLE

CHAPTER 2 — WORKED EXAMPLE

the derivative from first principles

THE DEFINITION

$$df/dx = [\, f(x + dx) - f(x) \,] \,/\, dx \qquad \text{as } dx \to 0$$

EXAMPLE $f(x) = x^2$

$$f(x + dx) = (x + dx)^2$$

$$= x^2 + 2x \cdot dx + (dx)^2$$

subtract $f(x) = x^2$:

$$= 2x \cdot dx + (dx)^2$$

divide by dx :

$$= 2x + dx$$

let $dx \to 0$:

$$df/dx = 2x$$

THE PARADOX

"rate of change at one INSTANT" — but change needs two points.

answer : it is what the slope APPROACHES as dx shrinks to nothing.

Why it matters for ML: gradient descent finds where the derivative is zero — the bottom of the loss valley. The sign tells the optimiser which way to step; the size tells it how big a step to take.

FORMULAS FROM THIS CHAPTER

Definition	$df/dx = [f(x+dx) - f(x)] / dx$, as $dx \to 0$
d/dx (x^2)	$2x$
Positive derivative	function increasing
Zero derivative	flat — peak, valley or plateau

TEST — Chapter 2

1. For $f(x) = x^2$, compute the derivative at $x = 3$ using $df/dx = 2x$. **→ 6**
2. Expand $(x + dx)^2$ and identify the term that survives after dividing by dx and letting $dx \to 0$. **→ $x^2 + 2x \cdot dx + (dx)^2$; the surviving term gives $2x$**
3. If $df/dx = 0$ at a point, what does that tell you about the graph there? **→ it is flat — a peak, valley, or plateau**

Derivative formulas through geometry

The power rule, and why it is true.

THE IDEA

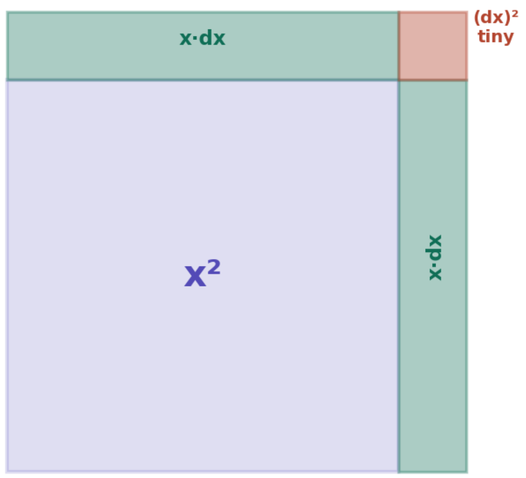
The standard derivative formulas are not arbitrary — each one can be seen geometrically. The most important is the **power rule**.

> **Power rule:** $d/dx\ (x^n) = n \cdot x^{n-1}$ — bring the power down in front, then subtract one from it.

WHY X^2 GIVES 2X

Picture a square of side x, area x^2. Grow the side by a tiny dx. Two new strips appear, each of area $x \cdot dx$, plus a tiny corner square $(dx)^2$ so small it can be ignored. So the growth is $2 \cdot x \cdot dx$; divide by dx and you get **2x**.

why $d(x^2)/dx = 2x$



growth = two strips = $2 \cdot x \cdot dx$ → divide by dx → 2x

THE RULES YOU NEED

Rule	Formula
Power	$d/dx\ (x^n) = n \cdot x^{n-1}$
Constant	$d/dx\ (c) = 0$
Constant multiple	$d/dx\ (c \cdot f) = c \cdot f'$
Sum	$d/dx\ (f + g) = f' + g'$
Trig	$d/dx\ (\sin x) = \cos x \cdot d/dx\ (\cos x) = -\sin x$

WORKED EXAMPLES

CHAPTER 3 — WORKED EXAMPLES

the formulas you will use every day

THE POWER RULE

$$\frac{d}{dx}(x^n) = n \cdot x^{n-1}$$

bring the power down in front, then subtract 1 from it

EXAMPLES

$\frac{d}{dx}(x^3) = 3x^2$

$\frac{d}{dx}(x^5) = 5x^4$

$\frac{d}{dx}(x) = 1$

$\frac{d}{dx}(7) = 0$ **(a constant never changes)**

CONSTANT MULTIPLE & SUM

$\frac{d}{dx}(5x^3) = 5 \cdot 3x^2 = 15x^2$

$\frac{d}{dx}(x^3 + x^2) = 3x^2 + 2x$

differentiate each term separately, then add

WORKED — $f(x) = 4x^3 + 2x^2 - 5x + 9$

$4x^3 \to 4 \cdot 3x^2 = 12x^2$

$2x^2 \to 2 \cdot 2x = 4x$

$-5x \to -5$

$9 \to 0$

$$\frac{df}{dx} = 12x^2 + 4x - 5$$

FORMULAS FROM THIS CHAPTER

Power rule	$\frac{d}{dx}(x^n) = n \cdot x^{n-1}$
Constant	$\frac{d}{dx}(c) = 0$
Constant multiple	$\frac{d}{dx}(c \cdot f) = c \cdot f'$
Sum	$\frac{d}{dx}(f+g) = f' + g'$
Trig	$\frac{d}{dx}(\sin x) = \cos x \cdot \frac{d}{dx}(\cos x) = -\sin x$

TEST — Chapter 3

1. Differentiate $f(x) = x^4$. **→ $4x^3$**
2. Differentiate $f(x) = 3x^2 + 5x - 7$. **→ $6x + 5$**
3. Differentiate $f(x) = 2x^3 - x^2 + 4x + 10$. **→ $6x^2 - 2x + 4$**

Visualizing the chain rule and product rule

The most important chapter for machine learning.

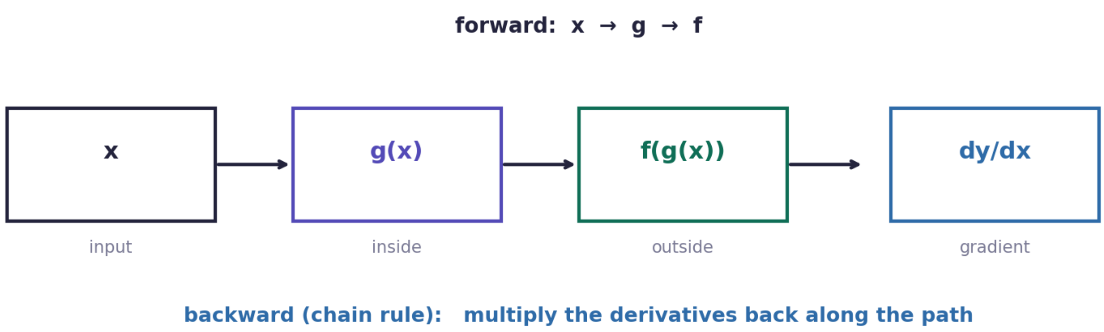
THE IDEA

Two rules let you differentiate combinations of functions: the **product rule** for things multiplied together, and the **chain rule** for functions nested inside functions.

Rule	Formula	Say it as
Product	$\frac{d}{dx}[f \cdot g] = f'g + fg'$	"first prime second, plus first second prime"
Chain	$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$	"derivative of the outside, times derivative of the inside"

THE CHAIN RULE, IN PICTURES

A nested function is a pipeline: x goes in, g acts on it, then f acts on that result. To find how the output responds to a nudge in x, you multiply the sensitivities along the pipeline.



WORKED EXAMPLES

CHAPTER 4 — WORKED EXAMPLES

the two rules that make backpropagation possible

PRODUCT RULE

$$\frac{d}{dx}[f \cdot g] = f' \cdot g + f \cdot g'$$

"first prime second, plus first second prime"

example $y = x^2 \cdot \sin(x)$

$f = x^2 \rightarrow f' = 2x$

$g = \sin x \rightarrow g' = \cos x$

$$dy/dx = 2x \cdot \sin(x) + x^2 \cdot \cos(x)$$

CHAIN RULE

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$$

"derivative of the OUTSIDE, times derivative of the INSIDE"

example 1 $y = (3x + 1)^5$

inside $g = 3x + 1 \rightarrow g' = 3$

outside $f = (\ )^5 \rightarrow f' = 5(\ )^4$

$$dy/dx = 5(3x + 1)^4 \cdot 3$$

$$= 15(3x + 1)^4$$

example 2 $y = \sin(x^2)$

inside $g = x^2 \rightarrow g' = 2x$

outside $f = \sin \rightarrow f' = \cos$

$$dy/dx = \cos(x^2) \cdot 2x$$

THIS IS BACKPROPAGATION. a neural network is functions inside functions — the chain rule is how the gradient travels back.

Why it matters for ML — this is the chapter that counts. A neural network *is* a deep nest of functions: layer inside layer inside layer. **Backpropagation is the chain rule** applied repeatedly, multiplying derivatives backward through the network. When gradients "vanish" or "explode", it is because those multiplied factors shrink toward zero or grow without bound.

FORMULAS FROM THIS CHAPTER

Product rule	$d/dx\,[f \cdot g] = f'g + fg'$
Chain rule	$d/dx\,[f(g(x))] = f'(g(x)) \cdot g'(x)$
In words	derivative of outside × derivative of inside
ML	backpropagation = repeated chain rule

TEST — Chapter 4

1. Differentiate $y = (2x + 5)^3$. **→ $3(2x+5)^2 \cdot 2 = 6(2x+5)^2$**
2. Differentiate $y = x^3 \cdot \cos(x)$. **→ $3x^2 \cdot \cos(x) - x^3 \cdot \sin(x)$**
3. Differentiate $y = \sin(4x^2)$. **→ $\cos(4x^2) \cdot 8x$**

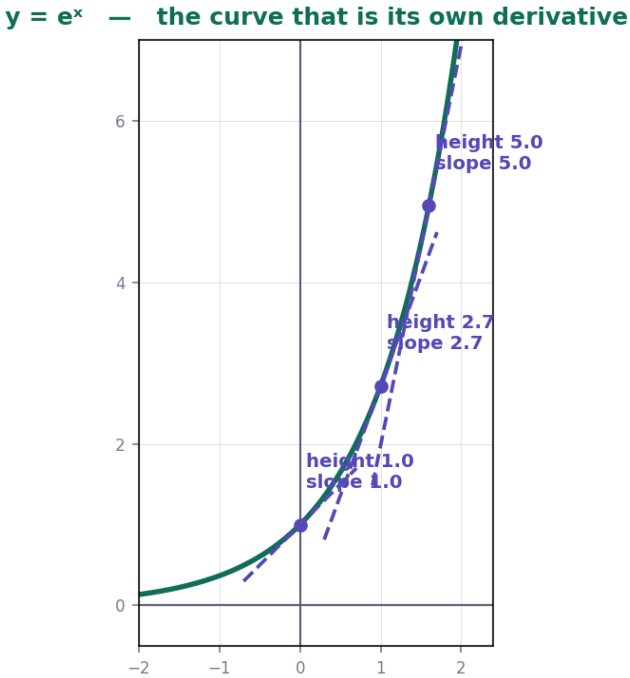
What's so special about Euler's number e?

The function that is its own derivative.

THE IDEA

Exponential functions have a special property: their rate of growth is proportional to their current size. Among all of them, exactly one has the growth rate **equal** to its size — the one with base **e** ≈ **2.71828**.

> **The defining fact:** $d/dx\,(e^x) = e^x$ — at every point, the height of the curve equals its slope.



THE FORMULAS

Function	Derivative
e^x	e^x
ln x	1 / x
a^x	$a^x \cdot \ln(a)$
e^(g(x))	e^(g(x)) · g′(x) (chain rule)

WORKED EXAMPLES

CHAPTER 5 — WORKED EXAMPLES

exponentials and the number e

THE KEY FACTS

$$\frac{d}{dx}(e^x) = e^x$$

the only function that is its own derivative

$$\frac{d}{dx}(\ln x) = 1/x$$

$$\frac{d}{dx}(a^x) = a^x \cdot \ln(a)$$

EXAMPLE 1 $y = e^{3x}$

chain rule : inside $g = 3x \rightarrow g' = 3$

$$dy/dx = e^{3x} \cdot 3 = 3e^{3x}$$

EXAMPLE 2 $y = e^{(x^2)}$

inside $g = x^2 \rightarrow g' = 2x$

$$dy/dx = e^{(x^2)} \cdot 2x = 2x \cdot e^{(x^2)}$$

EXAMPLE 3 $y = 2^x$

$$dy/dx = 2^x \cdot \ln(2)$$

ML link : e^x is inside the sigmoid and softmax functions, and ln x is inside cross-entropy loss.

> **Why it matters for ML:** e is everywhere in machine learning — inside the **sigmoid** function $1/(1+e^{-x})$, inside **softmax**, and its inverse ln appears in **cross-entropy loss**. Being able to differentiate these is how those losses get trained.

FORMULAS FROM THIS CHAPTER

d/dx (e^x)	e^x
d/dx (ln x)	$1/x$
d/dx (a^x)	$a^x \cdot \ln a$
With chain rule	$\frac{d}{dx} e^{g(x)} = e^{g(x)} \cdot g'(x)$

TEST — Chapter 5

1. Differentiate $y = e^{5x}$. **→ $5e^{5x}$**
2. Differentiate $y = e^{(3x^2)}$. **→ $e^{(3x^2)} \cdot 6x$**
3. Differentiate $y = \ln(x) + 3^x$. **→ $1/x + 3^x \cdot \ln(3)$**

Implicit differentiation, what's going on here?

When y cannot be separated from x.

THE IDEA

Sometimes y is tangled up with x and cannot be written as y = something. A circle, $x^2 + y^2 = 25$, is the classic case. The trick: differentiate **both sides** with respect to x, and remember that y is secretly a function of x — so every y picks up a **dy/dx** by the chain rule.

Term	Its derivative with respect to x
x^2	2x
y^2	**2y · dy/dx** — the chain rule adds the dy/dx
a constant	0
x·y	y + x·dy/dx — needs the product rule too

WORKED EXAMPLE

CHAPTER 6 — WORKED EXAMPLE

implicit differentiation

THE IDEA

when y is tangled with x and cannot be separated,

differentiate BOTH sides — and every y gets a dy/dx attached

EXAMPLE $x^2 + y^2 = 25$ (a circle)

differentiate each term with respect to x :

$x^2 \rightarrow 2x$

$y^2 \rightarrow 2y \cdot dy/dx$

$25 \rightarrow 0$

$$2x + 2y \cdot dy/dx = 0$$

$$2y \cdot dy/dx = -2x$$

$$dy/dx = -x / y$$

WHY y GETS dy/dx

because y is itself a function of x — this is the chain rule again.

THE METHOD IN THREE STEPS

1. Differentiate every term on both sides. **2.** Attach dy/dx to every term that came from a y. **3.** Collect the dy/dx terms and solve for dy/dx.

> **Why it matters for ML:** it is the same insight backpropagation relies on — variables that depend on other variables carry their derivatives with them through the chain rule.

FORMULAS FROM THIS CHAPTER

Method	differentiate both sides with respect to x
Every y	picks up a dy/dx
d/dx (y^2)	$2y \cdot dy/dx$
Then	collect dy/dx terms and solve

TEST — Chapter 6

1. Find dy/dx for $x^2 + y^2 = 100$. → **$-x/y$**
2. Find dy/dx for $y^2 = 4x$. → **$2y \cdot dy/dx = 4$ → $dy/dx = 2/y$**
3. Find dy/dx for $x^2 + 3y = 12$. → **$2x + 3 \cdot dy/dx = 0$ → $dy/dx = -2x/3$**

Limits, L'Hôpital's rule, and epsilon delta definitions

The rigorous foundation under everything above.

THE IDEA

Every derivative in this course was secretly a **limit** — the value a quantity approaches as dx goes to zero. This chapter makes that precise and gives you tools for the awkward cases.

> **A limit is what a function approaches**, not necessarily what it reaches. The function may not even be defined at that point — the limit can still exist.

THE THREE SITUATIONS

Case	What to do
Normal	Substitute the value directly
0/0 that factors	Factor, cancel the common term, then substitute
0/0 that resists	Use **L'Hôpital's rule**: differentiate top and bottom separately, then substitute

Warning: L'Hôpital's rule differentiates numerator and denominator *independently*. It is not the quotient rule, and it only applies to 0/0 or ∞/∞.

EPSILON–DELTA, IN ONE SENTENCE

The formal definition says: for any tolerance you demand on the output (epsilon), there is a closeness on the input (delta) that guarantees it. It is the rigorous way of saying "gets arbitrarily close".

WORKED EXAMPLES

CHAPTER 7 — WORKED EXAMPLES

limits and L'Hopital's rule

WHAT A LIMIT IS

the value a function APPROACHES — not necessarily reaches

EXAMPLE 1 simple substitution

$\lim\ (x^2 + 3)$ as $x \to 2$ $= 4 + 3 = 7$

EXAMPLE 2 the 0/0 case — factor it

$\lim\ (x^2 - 4) / (x - 2)$ as $x \to 2$

substituting gives 0/0 → undefined, so factor :

$(x - 2)(x + 2) / (x - 2) = x + 2$

$\to\ 2 + 2 = 4$

EXAMPLE 3 L'HOPITAL'S RULE

if you get 0/0 , differentiate top and bottom separately :

$\lim\ \sin(x) / x$ as $x \to 0$ gives 0/0

top $\sin x \to \cos x$

bottom $x \to 1$

$\to\ \cos(0) / 1 = 1$

note : differentiate them SEPARATELY — this is not the quotient rule.

Why it matters for ML: limits are why derivatives exist at all. They also explain practical failures — division by near-zero, log of near-zero, and the numerical-stability tricks (adding a small epsilon) you will see throughout ML code.

FORMULAS FROM THIS CHAPTER

Limit	the value approached, not necessarily reached
0/0 that factors	factor, cancel, substitute
L'Hôpital	differentiate top and bottom separately
Applies to	0/0 and ∞/∞ only

TEST — Chapter 7

1. Evaluate lim (3x + 1) as x → 4. **→ 13**
2. Evaluate lim $(x^2 - 9)/(x - 3)$ as x → 3. **→ factor: (x−3)(x+3)/(x−3) = x+3 → 6**
3. Use L'Hôpital on lim (e^x − 1)/x as x → 0. **→ top: e^x, bottom: 1 → $e^0/1 = 1$**

REFERENCE

Master formula sheet

Ch	Topic	Formula / rule
Ch 1	**Ring area**	$2\pi r \cdot dr$
Ch 1	**Circle area**	πR^2
Ch 1	**Integral**	add up infinitely many tiny pieces
Ch 1	**Derivative**	rate of change at an instant
Ch 2	**Definition**	$df/dx = [f(x+dx) - f(x)] / dx$, as $dx \to 0$
Ch 2	**d/dx (x^2)**	$2x$
Ch 2	**Positive derivative**	function increasing
Ch 2	**Zero derivative**	flat — peak, valley or plateau
Ch 3	**Power rule**	$d/dx\ (x^n) = n \cdot x^{n-1}$
Ch 3	**Constant**	$d/dx\ (c) = 0$
Ch 3	**Constant multiple**	$d/dx\ (c \cdot f) = c \cdot f'$
Ch 3	**Sum**	$d/dx\ (f+g) = f' + g'$
Ch 3	**Trig**	$d/dx\ (\sin x) = \cos x \cdot d/dx\ (\cos x) = -\sin x$
Ch 4	**Product rule**	$d/dx\ [f \cdot g] = f'g + fg'$
Ch 4	**Chain rule**	$d/dx\ [f(g(x))] = f'(g(x)) \cdot g'(x)$
Ch 4	**In words**	derivative of outside × derivative of inside
Ch 4	**ML**	backpropagation = repeated chain rule
Ch 5	**d/dx (e^x)**	e^x
Ch 5	**d/dx (ln x)**	$1/x$
Ch 5	**d/dx (a^x)**	$a^x \cdot \ln a$
Ch 5	**With chain rule**	d/dx e^g(x) = e^g(x) · g′(x)
Ch 6	**Method**	differentiate both sides with respect to x
Ch 6	**Every y**	picks up a dy/dx
Ch 6	**d/dx (y^2)**	$2y \cdot dy/dx$
Ch 6	**Then**	collect dy/dx terms and solve
Ch 7	**Limit**	the value approached, not necessarily reached
Ch 7	**0/0 that factors**	factor, cancel, substitute

Ch 7	**L'Hôpital**	differentiate top and bottom separately
Ch 7	**Applies to**	0/0 and ∞/∞ only

VERIFY HABITS

- **Chain rule:** name the inside and the outside *before* differentiating
- **Power rule:** the new power is always one less than the old one
- **Constants:** a lone number always differentiates to 0
- **Implicit:** every y must carry a dy/dx — check none were missed
- **L'Hôpital:** only for 0/0 or ∞/∞, and differentiate top and bottom separately

PERSONAL ERROR LOG — CARRIED FROM LINEAR ALGEBRA

Error	Fix
Misreading given numbers	Copy all values onto paper, labelled, before computing
Dropping a minus sign	Write subtraction as adding a negative
Assuming a shortcut applies	Check the condition first (as with eigenvalues on the diagonal)
Rushing the easy question	Same care on all three questions